

WEEK 11-15: INTEGRATIVE PHYSIOLOGY

GI SYSTEM

GENERAL PRINCIPLES AND ANATOMY

- Function: supply body with water, electrolytes, vitamins, and nutrients
- Layers: serosa, longitudinal & circular smooth muscle, submucosa, mucosa (lining)

GI MOTILITY AND ELECTRICAL ACTIVITY

1.) Gastrointestinal syncytium

- Gap junctions allow electrical signals to travel easily from cell to cell, causing the muscle layer to function as syncytium
- Travels in all directions

2.) Electrical waves — has two types

a.) Slow waves

- Not action potentials but are slow, undulating changes in the RMP
- Generated by interstitial cells of Cajal, which act as electrical pacemakers
- **Do not cause muscle contraction** (except possibly in the stomach)
 - Controls rhythm and appearance of spike potentials

b.) Spike potentials — true APs

- Occurs when the EMP becomes more positive than -40 mV
- Appear of the peaks of the slow waves; *higher slow wave peak = greater frequency of spike potentials*
- Causes muscle contraction

3.) Role of Calcium

- GI action potentials are caused by the slow entry of large numbers of Ca ions (some Na ions) through Ca-Na channels
- Entry of Ca during spike potentials that directly causes the muscle contraction

4.) MP changes

a.) Depolarization — less negative

- Makes the muscle more excitable
- Caused by:
 - Muscle stretching, ACh (parasympathetic), and GI hormones

b.) Hyperpolarization — more negative

- Makes the muscle less excitable
- Caused by:

- Norepinephrine/epinephrine and sympathetic stimulation

NEURAL CONTROL OF THE GI TRACT

1.) ENS

- GI tract's intrinsic nervous system, located within the gut wall
- Essential for controlling GI movement and secretion
- Consists of 2 main plexuses:

a.) Myenteric plexus — Auerbach's

- Located between muscle layers
- Controls GI movements
- Stimulation increases tone, contraction intensity, and rate

b.) Submucosal plexus — Meissner's

- Located in the submucosa
- Controls GI secretion and local blood flow

2.) Autonomic control — extrinsic

a.) Parasympathetic stimulation

- Travels via:
 - Vagus nerves (upper GI tract)
 - Sacral nerves (distal large intestine)
- Increases the activity of the entire ENS
- **ACh is the primary neurotransmitter**

b.) Sympathetic stimulation

- Originates from spinal segments T5-L2
- Inhibits GI tract activity due to the inhibitory effects of **norepinephrine** on the neurons of the ENS

3.) GI reflexes

a.) Reflexes entirely within ENS

- For local control of secretion, peristalsis, etc.

b.) Reflexes from the gut to sympathetic ganglia and back

- For long-distance signals (e.g., gastrocolic reflex)

c.) Reflexes from the gut to the SC & brain stem and back

- E.g., vagal, pain, and defecation reflexes

HORMONAL CONTROL OF MOTILITY

1.) Gastrin

- Secreted by G cells in the stomach

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- Stimulates gastric acid secretion (in response to food) and mucosal growth

2.) Cholecystokinin — CKK

- Secreted by I cells in the duodenum & jejunum in response to fat
- Strongly contracts the gallbladder (releases bile) and moderately inhibits stomach contraction

3.) Secretin

- Secreted by S cells in the duodenum in response to acid
- Promotes pancreatic bicarbonate secretion which neutralizes chyme

4.) Glucose-dependent insulintropic peptide — GIP

- Secreted by the upper small intestine
- Mildly decreases stomach motor activity and stimulates insulin secretion by the pancreas

5.) Motilin

- Secreted during fasting
- Increases GI motility and stimulates the interdigestive myoelectric complexes (motility waves every 90 minutes in a fasted state)

FUNCTIONAL MOVEMENTS

1.) Propulsive movements — peristalsis

- Basic propulsive movement; contractile ring forms and moves forward
- Stimulus: distension of the gut
- Requires an **active myenteric plexus**

LAW OF THE GUT

— myenteric reflex where a contractile ring forms behind (oral) the distended segment, and the gut relaxes downstream (anal), pushing the food toward the anus

IN SMALL INTESTINE

- Weak, propulsive waves; dies out after 3-5 cm
- Net movement is very slow (about 1 cm/min); chyme takes 3-5 hours to travel from the pylorus to the colon
 - **Gastroileal reflex**: stomach distension after a meal intensifies peristalsis in the ileum, pushing chyme into the large intestine
 - **Peristaltic rush**: abnormal, powerful, and rapid wave of peristalsis caused by intense

irritation (like infectious diarrhea) to quickly sweep contents into the colon

IN COLON AND DEFECATION — MASS MOVEMENTS

- Primary propulsive force in the colon; occurs 1-3 times daily (often after breakfast)
- A segment of the colon (often 20+ cm) contracts as a unit, propelling the fecal mass down the colon
- Triggered by **gastrocolic and duodenocolic reflexes**, which are initiated by stomach and duodenal distention

2.) Mixing movements — segmentation

- Keep intestinal contents mixed
- Achieved by peristaltic contractions blocked by a sphincter or by a local, intermittent segmentation contractions that “chop” the contents

IN SMALL INTESTINE

- Primary mixing movements, distinct from peristalsis
- Promotes mixing of chyme with digestive juices

IN COLON AND DEFECATION — HAUSTRATIONS

- “Segmentation” movements of the colon
- Combined contractions of the circular & longitudinal (teniae coli) muscles cause the colon to bulge into haustrations
- **“Digs and rolls”** the fecal matter, exposing all of it to the mucosal surface for fluid absorption

INGESTION AND SWALLOWING — DEGLUTITION

1.) Mastication — chewing

- Break down indigestible cellulose membrane in fruits and vegetables
- Increase the surface area of food for digestive enzymes
- Controlled by a chewing reflex initiated by food in the mouth

2.) Swallowing — deglutition

- Complex process divided into 3 stages:

a.) Voluntary stage

- Tongue consciously pushes the food bolus to the back of the pharynx

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b.) Pharyngeal stage

- Most critical stage and is an autonomic reflex; triggered when food stimulates receptors (especially on the tonsillar pillars)

- Airway protection: soft palate closes the nasal passages, vocal cords are pulled tightly together, epiglottis swings back to cover the larynx
- Propulsion: UES relaxes, fast peristaltic wave from the pharyngeal muscles forces the food into the esophagus

- Controlled by the swallowing center in the brain stem (medulla & pons) and briefly inhibits the respiratory centers

c.) Esophageal stage

- Transports food to the stomach

- Primary peristalsis: continuation of the swallowing wave from the pharynx (**takes 8-10 seconds to reach the stomach**)
- Secondary peristalsis: food remains in the esophagus, distension triggers these backup waves to clear the passage

- As the peristaltic wave travels, LES (prevents acid reflux in the esophagus by constricting) and upper part of the stomach relax ahead of the food

MOTOR FUNCTIONS

a.) Storage of food — 16 hours

- Food is stored in the body and fundus of stomach

- Vagovagal reflex (from stomach to the brain stem and back) reduces muscle tone in the stomach wall, allowing it to relax and accommodate 0.8 to 1.5 L of food with little increase in pressure

b.) Mixing and propulsion — with gastric secretion to form chyme

- Weak mixing waves begin in the midstomach and move toward antrum every 15-20 seconds

- These waves become much more powerful in the antrum, creating high-pressure constrictor rings

- Retropulsion: **most important mixing mechanism in the stomach**; as the wave hits the near-closed pylorus, most of the antral contents are squeezed backward

- Pyloric pump: intense peristaltic contractions in the antrum which **allows for “pumping” small amounts of chyme into the duodenum**

c.) Slowly emptying chyme into the small intestine

- Hunger contractions: intense rhythmic contractions that can fuse into a sustained contraction (lasting 2-3 minutes) when the stomach has been empty for many hours

- Associated with hunger pangs

REGULATION OF STOMACH EMPTYING

- Controlled by inhibitory signals from the duodenum, to ensure the small intestine is not overwhelmed

PROMOTING FACTORS

a.) Gastric food volume

- Stretching the stomach wall initiates myenteric reflexes that enhance the pyloric pump

b.) Hormone — gastrin

- Released in response to stomach stretch and protein, mildly enhances the activity of the pyloric pump

INHIBITORY FACTORS — DOMINANT CONTROL FACTORS

a.) Enterogastric nervous reflexes

- Strong reflexes triggered by duodenal distension, irritation, acidity (pH below 3.5-4), or an hypotonic/hypertonic chyme containing protein/fat breakdown products

- Inhibit the pyloric pump and increase the tone of the pyloric sphincter, slowing emptying

b.) Hormonal feedback

- Presence of fat in the duodenum is the most potent stimulus for inhibitory hormones

- CKK - **most potent hormone released due to fat**; inhibits stomach motility caused by gastrin
- Secretin - released due to acid
- GIP

MOVEMENTS OF THE COLON AND DEFECATION

- Function: proximal colon is for absorption of water and electrolytes; distal colon is for storage of feces; sluggish movements

- Ileocecal valve: protrudes cecum and, along with the ileocecal sphincter, **prevents backflow of feces from the colon into the small intestine**; cecal distension or irritation inhibits ileal emptying

DEFECATION

- Mass movements push feces into the rectum, triggering the urge to defecate

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- Only occurs when the external anal sphincter is voluntarily relaxed

ANAL SPHINCTERS

a.) Internal anal sphincter

- Involuntary smooth muscle

b.) External anal sphincter

- Voluntary striated muscle (*controlled by the pudendal nerve*)

DEFECATION REFLEXES

a.) Intrinsic reflex — weak

- Rectal distension activates the myenteric plexus; causes weak peristalsis and relaxing the internal sphincter

b.) Parasympathetic reflex — strong

- Rectal distension send signals to the sacral SC; send signals back via parasympathetic nerves that greatly intensify peristalsis and relax internal sphincter

GENERAL PRINCIPLE OF SECRETION — GLANDS

- Function of glands:

- Secreting digestive enzymes
- Secreting mucus for lubrication and protection

- Types of glands:

- **Mucous (goblet) cells:** single cells on the gut surface that secrete mucus in response to local irritation
- **Pits (crypts of Lieberkühn):** invaginations in the epithelium, found in the small intestine; contains specialized secretory cells
- **Tubular glands:** deep glands found in the stomach and upper duodenum (e.g., oxyntic glands)
- **Complex glands:** salivary glands, pancreas, and liver; located outside the alimentary tract wall and empty into it via ducts

BASIC STIMULATION OF GLANDS

1.) ENS

- Presence of food (via tactile stimulation, chemical irritation, or distention) activates the local ENS; stimulates gland secretion

2.) Parasympathetic stimulation

- Increases secretion rates, especially for the salivary glands, gastric glands, and pancreas

3.) Sympathetic stimulation

- Has dual effect:

- Slightly increase secretion on its own
- If secretion is already high (due to parasympathetic or hormonal drive), it reduces secretion by constricting BV to the glands

4.) Hormones

- Released from the GI mucosa into the blood and travel to glands to stimulate secretion

SALIVARY SECRETION

- Types of glands:

- Parotid (almost entirely serous fluid secretion), submandibular, sublingual

- Types of protein secretion from saliva:

- Serous secretion with ptyalin (alpha-amylase) for digesting starches
- Mucus secretion with mucin for lubrication

ESOPHAGEAL SECRETION

- Entirely mucous; provide lubrication for swallowing
- Compound glands near the stomach junction secrete mucus to protect the esophagus from acidic gastric reflux

KEY CELL TYPES OF OXYNTIC GLANDS

1.) Mucous neck cells

- Secrete mucus

2.) Peptic (chief) cells

- Secrete pepsinogen

3.) Parietal (oxyntic) cells

- Secretes HCl and intrinsic factor (for absorption of vitamin B12 in the ileum)

4.) Enterochromaffin-like (ECL) cells

- Secrete histamine

PHASES OF GASTRIC SECRETION

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1.) Cephalic stage

- Occurs before food enters the stomach; triggered by sight, smell, or thought of food; signal is transmitted by vagus nerve

2.) Gastric stage

- Occurs when food enters the stomach; triggers long vagovagal reflexes, local enteric reflexes, and the gastrin-histamine mechanism

3.) Intestinal stage

- Presence of food in the duodenum causes a small amount of continued gastric secretion

PANCREATIC SECRETION

- Secretes digestive enzymes from its acini and a large volume of sodium bicarbonate solution from its ducts

- Pancreatic digestive enzymes:

- **Protein:** trypsin, chymotrypsin, and carboxypolypeptidase
- **Carbohydrate:** pancreatic amylase (digest starch)
- **Fat:** pancreatic lipase, cholesterol esterase, and phospholipase

REGULATION OF PANCREATIC SECRETION

1.) ACh — from vagus

- Stimulates acinar cells to secrete digestive enzymes

2.) CCK

- Secreted by I cells in the duodenum in response to fats and proteins (peptones)
- Travels via blood and **strongly stimulates the acinar cells to secrete large amounts of digestive enzymes**

3.) Secretin

- Secreted by S cells in the duodenum in response to acid chyme (pH < 4.5)
- Travels via blood and strongly stimulates the ductal cells to secrete large volumes of sodium bicarbonate and H₂O

BILE SECRETION AND GALLBLADDER FUNCTION

- **Function of bile:**

- Fat digestion/absorption: bile acids (salts) emulsify large fat globules and form **micelles**, essential for “ferrying” digested fats to intestinal mucosa for absorption
- Excretion: bile is the **primary route for excreting waste products like bilirubin** (from hemoglobin destruction) and **excess cholesterol** (gallstone)

- **Secretion and storage:**

- Secreted in 2 stages:
 - **Hepatocytes** secrete the primary biles (bile acids, cholesterol)
 - Ducts add a watery, bicarbonate-rich solution (stimulated by secretin)
- Bile is continuously secreted by the liver but is stored and concentrated in the gallbladder (5- to 20-fold by actively absorbing H₂O, Na, Cl)

EMPTYING OF THE GALLBLADDER

- CCK is the most potent stimulus for gallbladder emptying

- Released from the duodenum when fatty food enter

- Causes 2 simultaneous events:

- Contraction of bladder
- **Relaxation of the sphincter of Oddi**, which allows bile to flow into the duodenum

ENTEROHEPATIC CIRCULATION OF BILE SALTS

- About 94% of bile salts are reabsorbed from the small intestine (especially the distal ileum) back into the portal blood

- Liver re-absorbs these salts and re-secretes them

- This is crucial as the body's small pool of bile salts cycles ~17 times per day to produce enough bile

GALLSTONES

- Cholesterol precipitation of a solution forms cholesterol gallstones from high-fat diet or gallbladder inflammation

- **Causes:**

- Too much reabsorption of H₂O and bile acids from bile, too much cholesterol in bile, and inflammation of epithelium

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SECRETIONS OF THE SMALL INTESTINE

1.) Brunner's glands — duodenum

- Secrete large amounts of alkaline mucus
- Protect the duodenal wall from the highly acidic gastric juice
- Inhibited by sympathetic stimulation, which may be a factor in duodenal ulcers

2.) Crypts of Lieberkühn — entire small intestine

- Enterocytes in the crypts secrete a large volume of watery fluid (an alkaline, extracellular fluid-like solution)
- **Acts as a “watery vehicle”** to help absorb nutrients from the chyme

3.) Digestive enzymes

- Located within the enterocytes that cover the villi
- Digest food as it is being absorbed and include:
 - Peptidases, disaccharidases (sucrase, maltase, lactase), and intestinal lipase

SECRETIONS OF THE LARGE INTESTINE

- Mucosa of large intestine has many crypts of Lieberkühn but contain no villi
- Epithelial cells secrete almost no digestive enzymes
- **Primary secretion is mucous**
 - Protects intestinal walls from excoriation and bacterial activity
 - Provides an “adherent medium” for holding fecal matter together
 - Contains bicarbonate, providing an alkaline (pH 8.0) barrier against acids formed by bacteria in the feces (foul smell)

DIGESTION OF CARBOHYDRATES

PRIMARY SOURCES

- Lactose (milk sugar), sucrose (table sugar), and starches (polysaccharides)

IN THE MOUTH AND STOMACH

- Digestion begins in the mouth with the enzyme ptyalin (salivary-alpha amylase), which starts breaking down starches
- Continues in the stomach for up to an hour until stomach acid (pH < 4.0) inactivates the enzyme

IN THE SMALL INTESTINE — PANCREATIC

- Pancreatic alpha-amylase (more powerful) breaks down the remaining starches into maltose and other small glucose polymers

AT THE BRUSH BORDER

- Final breakdown occurs in the intestinal brush border
- Enzymes located in the membranes of the enterocytes (intestinal cells) split disaccharides into monosaccharides:
 - Lactase splits lactose -> glucose + galactose
 - Sucrase splits sucrose -> glucose + fructose
 - Maltase & alpha-dextrinase split maltose/glucose polymers -> glucose

FINAL PRODUCT

- Monosaccharides (glucose, galactose, fructose)
 - *Glucose* accounts for over 80% of the final product

DIGESTION OF PROTEINS

IN THE STOMACH

- Pepsin (only functions in a highly acidic environment) begins protein digestion
- Important in digesting collagen
- Break proteins down into smaller polypeptides (10-20% of total digestion)

IN THE SMALL INTESTINE — PANCREATIC

- Most protein digestion occurs in the duodenum and jejunum
- Pancreatic enzymes (trypsin, chymotrypsin, carboxypolypeptidase, and elastase) break large polypeptides into smaller tripeptides and dipeptides

AT THE BRUSH BORDER

- Peptidases in the enterocyte membrane split the peptides into dipeptides, tripeptides, and a few single amino acids

INSIDE THE CELL

- Dipeptides and tripeptides are transported into the intestinal cell, where intracellular peptidases split them into their final form

WEEK 11-15: INTEGRATIVE PHYSIOLOGY

FINAL PRODUCT

- Over 99% of protein is absorbed as individual amino acids

DIGESTION OF FATS

PRIMARY SOURCE

- Most abundant dietary fats are triglycerides (neutral fats)

STEPS

1st: emulsification (in the duodenum)

- Critical first step
- Large fat globules are broken down into tiny droplets by agitation, made possible by bile salts and lecithin (acts as detergents)
- Emulsification massively increases the surface area for enzymes to work on

2nd: hydrolysis (in the small intestine)

- Pancreatic lipase (most important fat-digesting enzyme) attacks the surface of the emulsified droplets and hydrolyzes the triglycerides

3rd: micelle formation

- Bile salts form micelles that have a fat-soluble center and water-soluble exterior
- Essential because they “ferry” these insoluble fat products through the watery chyme to the surface of the intestinal cells, where they can be absorbed

FINAL PRODUCTS

- Free fatty acids and 2-monoglycerides

Fat -> emulsified fat -> fatty acids and 2-monoglycerides

2.) Protein — amino acids

- Absorbed (as dipeptides, tripeptides, or amino acids) via Na co-transport (secondary active transport, similar to glucose)
- Any peptides that enter the cell are broken into single amino acids before entering the blood

3.) Fats

- Micelles bring fatty acids and monoglycerides to the cell surface, where they diffuse directly across the cell membrane (because they are lipid-soluble)

a.) Inside the cell

- Fatty acids and monoglycerides are taken to the SER and re-synthesized back into triglycerides

b.) Chylomicron formation

- New triglycerides are packaged with cholesterol and proteins into large globules

c.) Entry into circulation

- Chylomicron are released from the cell base and enter the lymphatic system via the central lacteal of the villus, bypassing the liver

d.) Exception

- Short- and medium-chain fatty acids (e.g., from butterfat) are water-soluble and are absorbed directly into the portal blood, not the lymph

ABSORPTION IN THE LARGE INTESTINE — FECES FORMATION

- Function: proximal half (absorbing colon) absorbs water and electrolytes; distal half (storage colon) stores feces

a.) Absorption

- Large intestine absorbs Na actively and H₂O passively (via osmosis)
- Tight junctions allows to absorb Na against a very high concentration gradient and preventing ion back-leak

b.) Bacterial action: colon bacilli are present and perform several functions:

- Digest small amounts of cellulose (negligible energy for humans)
- Produce vitamin K (for blood coagulation and often insufficient in the diet)
- Produce vitamin B₁₂, thiamine, and flatus gases

ABSORPTION OF NUTRIENTS — SMALL INTESTINE

1.) Carbohydrates — monosaccharides

a.) Glucose and galactose

- Absorbed via secondary active transport, powered by the Na gradient
- Transported by the SGLT1 protein co-transports one Na with one glucose/galactose molecule into the cell

b.) Fructose

- Absorbed by facilitated diffusion via GLUT5 transporter
- This process does not require Na and is slower

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c.) Feces

- Normally $\frac{3}{4}$ H₂O and $\frac{1}{4}$ solid
- Solid matter consists of dead bacteria, undigested roughage, fat, and inorganic matter
- The brown color is from stercobilin and urobilin (bilirubin derivatives)

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ENDOCRINE SYSTEM

THREE CHEMICAL CLASSES OF HORMONES

1.) Protein and polypeptides

- Majority of hormones in the body; water-soluble
- Synthesized in the ER and packaged in the golgi into secretory vesicles
- Examples: pituitary gland, insulin, glucagon, and GH

2.) Steroids

- Synthesized from cholesterol and are not stored in significant amounts
- Examples: cortisol, aldosterone, estrogen, progesterone, and testosterone

3.) Amines — tyrosine derivatives

- Derived from the amino acid tyrosine
- Examples: thyroid hormones (thyroxine and triiodothyronine) and adrenal medullary hormones (epinephrine and norepinephrine)

HORMONE CONTROL AND TRANSPORT

1.) Negative feedback

- Where hormone secretion is tightly controlled
- Most common control system
- After a hormone is released, the effects of the hormone (or the products it generates) tend to suppress its own further release, preventing overactivity

2.) Positive feedback

- Rare; hormone's action causes additional secretion of the hormone
- E.g., surge of LH caused by estrogen before ovulation

CYCLICAL VARIATIONS

- Hormone release is often cyclical, influenced by the daily (diurnal) cycle, sleep, and aging

TRANSPORT IN BLOOD

- Dependent on its chemical class

a.) Water-soluble hormones — peptides and catecholamines

- Dissolve directly in the plasma and are transported to their target tissues
- Cleared from the blood quickly, often having a **half-life of only minutes**

b.) Steroid and thyroid hormones

- Lipid-soluble and circulate in the blood primarily bound to plasma proteins
- More than 99% of thyroxine is bound
 - Biologically inactive: while bound, these hormones are inactive
 - Reservoir: protein-bound hormones **act as a reservoir**, slowly releasing free hormones to the target tissues
 - Slow clearance: binding greatly slows clearance from the blood, giving them a **long half-life of hours to days**

MECHANISM OF HORMONE ACTION — RECEPTORS

- Hormone's action begins by binding to a specific receptor; location of the receptor depends of the hormone type

a.) Receptors in/on the cell membrane

- Used by protein, peptide, and catecholamine hormones

b.) Receptor in the cytoplasm

- Primary receptors for steroid hormones

c.) Receptors on the nucleus

- Receptors for thyroid hormones

- Target cells can adjust their sensitivity to a hormone by changing the number of active receptors

a.) Down-regulation

- Increase in hormone concentration can cause the number of active receptors to decrease, making the tissue less responsive

b.) Up-regulation

- Hormone can induce the formation of more receptors, making the tissue progressively more sensitive

INTRACELLULAR SIGNALING AND SECOND MESSENGERS

- When hormone binds its receptor, it initiates a cascade of reactions

a.) Action via intracellular receptors — steroid & thyroid hormones

- Enter the cell, bind to their receptor in the cytoplasm or nucleus, and the hormone-receptor complex directly interacts with the cell's DNA

b.) Gene transcription

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- The complex binds to a specific hormone response element on the DNA

c.) Protein synthesis

- Binding activates or represses the transcription of specific genes, leading to the formation of new proteins (like enzymes) that carry out the cell's response

d.) Slow response

- Slow because it requires synthesizing new proteins; effects may be delayed for minutes, hours, or even days

ACTION VIA MEMBRANE RECEPTORS — PEPTIDE & CATECHOLAMINES

- These hormones **act as the first messenger** by binding to a surface receptor
- Binding triggers the creation of an intracellular second messenger that causes the cell's response

a.) G protein-linked receptors

- Activate an intracellular G protein, which activates a second messenger system

b.) Adenylyl cyclase-cAMP system

- The hormone-receptor complex (via a Gs protein) activates adenylyl cyclase
- This enzyme converts ATP into the second messenger, cAMP
- cAMP then activates cAMP-dependent protein kinase, which phosphorylates proteins, leading to a cascade of effects and the final cell response

c.) Phospholipase C system

- The hormone-receptor complex activates phospholipase C
- Splits a membrane phospholipid (PIP₂) into 2 different second messenger: inositol triphosphate (IP₃) and diacylglycerol (DAG)
 - IP₃ mobilizes Ca ions from the ER
 - DAG activates the enzyme protein kinase C, which phosphorylates proteins

d.) Calcium-calmodulin system

- Ca ions (released by IP₃ or by channels opening) act as a second messenger
- Ca binds to protein calmodulin
- Ca-calmodulin complex then activates or inhibits other protein kinases to cause the cell's response

THE PITUITARY GLAND: ANT. VS POST.

- Pituitary gland (hypophysis) is divided into 2 distinct lobes: anterior (adenohypophysis) and posterior (neurohypophysis) pituitary

- **Anterior:** synthesizes and secretes 6 hormones
- **Posterior:** does not synthesize hormones; only stores and secretes two hormones that are produced by the hypothalamus

HYPOTHALAMIC CONTROL OF THE PITUITARY

- Hypothalamus is the primary control center for all pituitary secretion, but uses 2 different methods:

a.) Control of the anterior pituitary

- Secretes releasing and inhibitory hormones into a specialized BVs system (**hypothalamic-hypophyseal portal vessels**)
- These portal vessels carry the hormones directly to the anterior pituitary, where they act on glandular cells to control their secretion

b.) Control of the posterior pituitary

- Controlled by nerve signals
- Large neurons (magnocellular neurons), located in the supraoptic and paraventricular nuclei of the hypothalamus, send their axons down the pituitary stalk
- Hormone-releasing nerve impulses travel down these axons and cause the hormones to be released directly from the nerve endings

ANTERIOR PITUITARY HORMONES

CELLS — PRODUCES ITS HORMONES

- a.) Gonadotropes: secrete LH and FSH
- b.) Lactotropes: secrete prolactin (PRL)
- c.) Thyrotropes: secrete thyroid-stimulating hormone (TSH)
- d.) Somatotropes (30-40% of cells): secrete GH
- e.) Corticotropes (20% of cells): secrete adrenocorticotrophic hormone (ACTH)

HORMONES

- a.) GNRH (growth hormone-releasing hormone): stimulates GH
- b.) GHIH or somatostatin: inhibits GH
- c.) CRH (corticotropin-releasing hormone): stimulates ACTH
- d.) TRH (thyrotropin-releasing hormone): stimulates TSH

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e.) GnRH (Gonadotropin-releasing hormone): stimulates LH and FSH
f.) PIH and dopamine: inhibits prolactin

GROWTH HORMONE

- Unique because it affects almost all tissues of the body, rather than just a specific target gland
- Overall effect is to enhance body protein, decrease fat stores, and conserve carbohydrates
- Stimulates the growth of the skeletal frame
- Causes long bones to grow in length at the epiphyseal cartilages (in children)
- Causes bone to grow in thickness by stimulating osteoblasts

GROWTH HORMONE: REGULATION & ABNORMALITIES

REGULATION

- GH secretion is pulsatile; strongly stimulated by:
 - Starvation (especially in protein deficiency)
 - Hypoglycemia (low blood sugar)
 - Exercise, trauma, or excitement
 - The first two hours of deep sleep

PANHYPOPITUITARISM — DWARFISM

- Deficiency of all anterior pituitary hormones during childhood
- Person's body parts develop proportionally but at a greatly decreased rate

GIGANTISM

- Caused by excess GH (from a tumor) before adolescence, while the epiphyseal plates are still open

ACROMEGALY

- Caused by excess GH after adolescence
- The person cannot grow taller, but their bones grow thicker, especially the hands, feet, jaw, and cranium
- Causes soft tissue organs to enlarge

POSTERIOR PITUITARY HORMONES

a.) Antidiuretic hormone (ADH/vasopressin)

- Synthesis: primarily synthesized in the supraoptic nucleus of the hypothalamus

- Function: controls water conservation in the kidneys
- Mechanism: ADH makes the kidney's collecting ducts and tubules *permeable to water*; stimulates the insertion of aquaporins (water pores) into the cell membranes

- **With ADH**: H₂O is reabsorbed (conserved), and urine is concentrated
- **Without ADH**: H₂O is lost, and urine is very diluted

- Regulation:

- Increased osmolarity: most powerful stimulus; osmoreceptors in the hypothalamus detect that the blood is too concentrated (salty), triggering ADH release to retain water
- Low blood volume/pressure: a large drop in blood volume (sensed by atrial and baroreceptors) strongly stimulates ADH release

- Vascular effect: at high concentrations, *ADH is a potent vasoconstrictor and raises BP* (why it is also called vasopressin)

b.) Oxytocin

- Synthesis: primarily synthesized in the paraventricular nucleus of the hypothalamus

- Function 1: Milk ejection

- Best-understood role; sucking stimulus on the nipple sends a nerve signal to the hypothalamus, which causes the posterior pituitary to release oxytocin
- Oxytocin travels to the breast and causes the *contraction of myoepithelial cells*, which ejects milk from the alveoli

- Function 2: uterine contraction

- Oxytocin powerfully stimulates contraction of the pregnant uterus, especially near the end of gestation, and is believed to be at least partially responsible for labor and birth

THYROID HORMONE SYNTHESIS, STORAGE, AND RELEASE

- Hormones: thyroxine (T₄) and triiodothyronine (T₃)

- Production rate: about 93% is T₄, and 7% is T₃

- Potency: T₃ is 4X as potent as T₄; in most body tissues, most of T₄ is eventually converted into T₃ (primary active hormone)

- Core structure: composed of follicles filled with colloid; **its major component is thyroglobulin (contains tyrosine amino acids)**

- Unique storage: thyroid can store large amounts of hormone within the thyroglobulin molecules, holding a 2 to 3-month supply for the body

WEEK 11-15: INTEGRATIVE PHYSIOLOGY

TRANSPORT, MECHANISM, AND DURATION OF ACTION

- Transport in blood: over 99% of T4 and T3 in the blood are **bound to plasma proteins** (mainly thyroxine-binding globulin)
- Slow onset & long duration: this high-affinity binding causes the hormones to be released to the tissues very slowly
 - T4: has a **long latent period of 2-3 days**, with maximal activity not seen for 10-12 days; its half-life is about 15 days
 - T3: acts about 4X faster, with a **latent period of 6-12 hours** and maximal activity in 2-3 days
- Mechanism of action (genomic): activates nuclear transcription of genes
 - **T4 functions mainly as a prohormone**; inside the target cell, T4 is converted to T3 by removing one iodide
 - T3 then binds to an intracellular thyroid hormone receptor located in the nucleus, often **forming a heterodimer with the retinoid X receptor (RXR)**
- Nongenomic actions: thyroid hormones can also produce rapid, non-genomic effects at the plasma membrane and mitochondria

DISEASES OF THE THYROID GLAND

1.) Hyperthyroidism — overactive thyroid

- Symptoms: heat tolerance, weight loss, increased sweating, diarrhea, muscle weakness, nervousness, fatigue but inability to sleep, and a fine muscle tremor
- Examples: Grave's disease, thyroid adenoma

2.) Hypothyroidism — underactive thyroid

- Symptoms: fatigue, extreme somnolence, muscular sluggishness, slower HR, weight gain, constipation, mental sluggishness, husky voice, and increased blood cholesterol (leads to atherosclerosis)
- Examples: Hashimoto's disease, endemic colloid goiter, myxedema, cretinism

ADRENAL CORTEX

- Three distinct layers:

a.) *Zona glomerulosa (15% of cortex)*

- Outer layer; secrete large amounts of aldosterone (primary mineralocorticoid) because of aldosterone synthase

- **Secretion is controlled by angiotensin II and K**

b.) *Zona fasciculata (75% of cortex)*

- Middle layer; secretes glucocorticoids cortisol (95% of all glucocorticoid activity) and corticosterone, as well as small amounts of androgens

- Secretion is **controlled by ACTH (adrenocorticotrophic hormone)** from the pituitary gland

c.) *Zone reticularis*

- Inner layer; secretes adrenal androgens (like DHEA and androstenedione) and small amounts of glucocorticoids

- Secretion is also **controlled by ACTH**

FUNCTIONS OF MINERALOCORTICIDS — ALDOSTERONE

- Loss of mineralocorticoids is lethal, causing severe Na loss, hyperkalemia (high K), and circulatory shock

PRIMARY RENAL EFFECT

- Increase Na reabsorption and increase potassium secretion by the principal cells of the renal collecting tubules

EFFECT ON ECF AND BP

- Excess aldosterone causes the body to retain Na and H₂O, increasing ECF volume and arterial pressure
- "**Aldosterone escape**": after 1-2 days of high aldosterone, the high BP causes pressure natriuresis (increased Na excretion), which returns Na output to normal. This "escape" prevents lethal fluid retention, but the hypertension persists

EFFECT ON K

- Excess aldosterone: causes hypokalemia (low K) by secreting it into the urine; **lead to severe muscle weakness**
- **Aldosterone deficiency**: causes hyperkalemia (high K), which can lead to severe **cardiac toxicity and heart failure**

ACID-BASE BALANCE

- Excess aldosterone increases H⁺ ion secretion in the kidneys, leading to a mild metabolic alkalosis

WEEK 11-15: INTEGRATIVE PHYSIOLOGY

OTHER EFFECTS

- Conserves Na in sweat gland, salivary glands, and intestines

FUNCTIONS OF GLUCOCORTICOIDS — CORTISOL

- 1.) Essential for resisting physical and mental stress
- 2.) Best-known effect is stimulating gluconeogenesis in the liver
- 3.) Decreases glucose utilization by most cells
- 4.) Decrease protein stores in all body cells except liver
- 5.) Can lead to muscle weakness and a suppressed immune system
- 6.) Increases liver protein and plasma protein production
- 7.) Promotes the mobilization of fatty acids from adipose tissue, increasing their use of energy
- 8.) Any type of stress (trauma, infection, surgery, etc.) causes the release of ACTH, followed by a large secretion of cortisol
- 9.) High levels of cortisol can block or reverse inflammation
- 10.) Decreases the number of eosinophils and lymphocytes in the blood
- 11.) Increases the production of RBC
- 12.) “Adrenal diabetes”: combined effect of increased gluconeogenesis and decreased glucose utilization raises blood glucose levels
- 13.) Peculiar obesity: excess cortisol causes a unique type of obesity, depositing fat in the chest and head, leading to a “**buffalo-like torso**” and a “**moon face**”

ADRENOCORTICAL ABNORMALITIES

1.) Hypoadrenalism — Addison’s disease

- Inability of the adrenal cortices to produce hormones, often due to autoimmune atrophy

2.) Hyperadrenalism — Cushing’s syndrome

- Excess cortisol

3.) Primary aldosteronism — Conn’s syndrome

- Caused by tumor in the zona glomerulosa secreting large amounts of aldosterone

4.) Adrenogenital syndrome

- An adrenal tumor secretes excessive androgens

PANCREAS ANATOMY AND HORMONES

- Contains islets of Langerhans, which secrete hormones directly into the blood
- Islets 3 major cell types:
 - Beta cells (60%): secrete insulin and amylin
 - Alpha cells (25%): secrete glucagon
 - Delta cells (10%): secrete somatostatin
- **Regulate each other**: insulin inhibits glucagon, and somatostatin inhibits both

INSULIN: THE HORMONE OF ENERGY ABUNDANCE

- Secretion is associated with energy abundance; when a diet is rich in energy, especially carbohydrates, insulin is secreted
- Insulin’s primary role is to **store excess energy**:
 - Causes excess carbohydrates to be stored as glycogen in the liver and muscles
 - Any carbohydrates that cannot be stored as glycogen are converted by insulin into fats
 - Promotes the uptake of amino acids and their conversion into protein, while also inhibiting the breakdown of existing proteins

CELLULAR EFFECTS OF INSULIN

a.) Rapid effect (seconds): markedly increases the uptake of glucose in about 80% of the body’s cells, especially muscle and adipose tissue

- Achieved by causing the translocation of intracellular vesicles containing glucose transporters (like GLUT4) to the cell membrane
- Brain is major exception: most brain cells do not require insulin for glucose uptake

b.) Slower effects (hours-days): promotes protein synthesis and changes gene transcription for metabolic enzymes

INSULIN’S EFFECT

- 1.) After a meal (high insulin), the liver takes up and stores glucose as glycogen
- 2.) **Inhibits gluconeogenesis**
- 3.) Insulin promotes fat storage (anabolic)
- 4.) Insulin deficiency promotes fat use (catabolic)
- 5.) Insulin promotes protein storage (anabolic)

WEEK 11-15: INTEGRATIVE PHYSIOLOGY

GLUCAGON: THE “HYPERGLYCEMIC” HORMONE

- Secreted by the alpha cells and has functions diametrically opposed to insulin
- Main function is to **increase blood glucose concentration**
- Primary effects:
 - **Glycogenolysis:** breakdown of liver glycogen (most dramatic effect)
 - **Gluconeogenesis:** increase the synthesis of glucose from amino acids in the liver
- Mechanism of action:
 - Glucagon **activates a cAMP** second messenger cascade
 - This cascade system provides amplifying mechanisms, where a tiny amount of glucagon can cause a large release of glucose
- Regulation of secretion:
 - **Low blood glucose:** stimulates glucagon
 - **High blood glucose:** inhibits glucagon
 - **High amino acids:** stimulates glucagon; converts amino acids into glucose
 - **Exercise:** stimulates glucagon

SOMATOSTATIN: THE INHIBITOR

- Secreted by delta cells in response to food ingestion (increased glucose, amino acids, and fatty acids)
- **It is an inhibitory hormone:**
 - Acts locally in the islets to depress the secretion of both insulin and glucagon
 - Decreases GI motility, secretion, and absorption
- **Role:** “extend” the time over which food nutrients are assimilated into the blood

BLOOD GLUCOSE REGULATION SUMMARY

- Body maintains a narrow range of blood glucose (fasting 80-90 mg/100 mL)

WHY IS THIS IMPORTANT?

- a.) Hypoglycemia: the brain, retina, and gonads depend almost exclusively on glucose for energy and do not require insulin for its uptake
- b.) Hyperglycemia: high glucose causes cellular dehydration (due to osmotic pressure), loss of glucose in the urine (osmotic diuresis), and long-term tissue and vascular injury (leading to heart attack, stroke, kidney disease, and blindness)

KEY REGULATORY SYSTEMS

a.) Liver

- Acts as a “glucose buffer system,” storing glycogen when glucose is high and releasing it when glucose is low

b.) Insulin-glucagon system

- Primary feedback mechanism

c.) Hypothalamus/epinephrine

- Severe hypoglycemia stimulates the sympathetic NS to release epinephrine, which causes rapid glycogenolysis

d.) GH and cortisol

- Secreted during prolonged hypoglycemia to decrease glucose utilization and increase fat utilization

DIABETES MELLITUS

- A syndrome of impaired carbohydrate, fat, and protein metabolism

a.) Type 1 diabetes — *insulin-dependent*

- Cause: **lack of insulin secretion** due to autoimmune destruction of pancreatic beta cells
- Characteristic: usually occurs in **younger people** (<20 years); plasma insulin is low or absent
- Treatment: insulin administration

b.) Type 2 diabetes — *non-insulin dependent*

- Cause: **insulin resistance**, a decreased sensitivity of target tissues to insulin
- Characteristics: far more common (90-95% of cases); strongly associated with obesity, especially abdominal fat; plasma insulin is often high (hyperinsulinemia) as the pancreas compensates for the resistance
- Treatment: lifestyle changes (weight loss, exercise), oral medication

HYPERINSULINISM — INSULINOMA

- A rare condition caused by an **insulin-secreting adenoma (tumor) in the islets of Langerhans**
- Causes severe hypoglycemia and “insulin shock”
- Symptoms:
 - **< 20 mg/100 mL:** coma
 - **20-50 mg/100 mL:** clonic seizures and loss of consciousness

WEEK 11-15: INTEGRATIVE PHYSIOLOGY

- **50-70 mg/100 mL:** nervousness, CNS excitability, sweating, trembling
- Treatment: immediate IV administration of glucose or glucagon

CALCIUM & PHOSPHATE: THE BASICS

- Extracellular Ca ion concentration is one of the most tightly regulated systems in the body (**normal: 9.4 mg/dL**)
- PO is less tightly regulated
- The bone reservoir: the vast majority of the body's Ca (99%) and phosphate (85%) is stored in the bones; it acts as a large reservoir to buffer plasma Ca levels
- Calcium is essential for muscle contraction, blood clotting, and nerve impulse transmission
 - **Hypocalcemia:** causes nervous system excitement and tetany (spontaneous muscle contraction); key sign is **carpopedal spasm**
 - **Hypercalcemia:** depresses the nervous system and the muscle activity, leading to **slugging reflexes, constipation, and lack of appetite**

3 KEY HORMONES OF CA CONTROL

1.) PTH

- Most important hormone for Ca regulation
- Its net effect is to increase plasma Ca and decrease plasma PO

- Source: secreted by chief cells of the 4 parathyroid glands
- Primary stimulus:
 - Secretion is stimulated by low plasma Ca
 - Parathyroid cells detect this using Ca-sensing receptors (CSRs)
 - High Ca inhibits PTH secretion

KEY ACTION

- a.) On bone (resorption):** PTH stimulates the release of Ca and PO from bones; two phases:
 - Rapid phase (osteolysis): a fast release of Ca from the "exchangeable" pool in the bone fluid, managed by osteocytes
 - Slow phase (osteoclast activation): PTH stimulates osteoblasts, which in turn release RANKL to activate osteoclasts

b.) On kidneys:

- Increases the reabsorption of Ca in the distal

- tubules, preventing it from being lost in urine
- Strongly increases the excretion of PO, which is why it lowers plasma PO

c.) On intestine (indirect):

- Critical step; PTH stimulates the kidneys to perform the final activation step of vitamin D
- The newly activated vitamin D increases Ca absorption in the intestine

2.) Vitamin D — active form

- Increase Ca reabsorption from the intestine
- Not biologically active in its initial form and must be activated in a two-step process
- **Final, critical kidney step is stimulated by PTH**

ACTIVATION PATHWAY

- a.) Skin:** 7-dehydrocholesterol is converted to cholecalciferol (vitamin D3) by UV light from the sun
- b.) Liver:** D3 is converted to 25-hydroxycholecalciferol; main storage form
- c.) Kidney:** 25-hydroxycholecalciferol is converted to 1,25-dihydroxycholecalciferol; final, most active form of vitamins D

PRIMARY ACTION

- Active vitamin D functions as a hormone to promote intestinal Ca absorption
- It does this by increasing the synthesis of calbindin, a Ca-binding protein, within the intestinal epithelial cells
- It also promotes intestinal phosphate absorption

3.) Calcitonin

- Minor hormone that weakly decreases plasma Ca
- Has very weak effect on plasma Ca in adult humans
- **PTH system is dominant**
- Its effects are much more significant in children, who have a high rate of bone remodeling

- Source: secreted by the parafollicular cells (C cells) of the thyroid glands
- Primary stimulus: secretion is stimulated by high plasma Ca (opposite of PTH)
- Action: it decreases plasma Ca by inhibiting the activity of osteoclasts

PATHOPHYSIOLOGY OF CALCIUM AND BONE

1.) Hypoparathyroidism

WEEK 11-15: INTEGRATIVE PHYSIOLOGY

- Leads to hypocalcemia and tetany

2.) Primary hyperparathyroidism

- Caused by a parathyroid tumor
- Leads to hypercalcemia, hypophosphatemia (low phosphate), severe bone resorption (osteitis fibrosa cystica), and a high risk of kidney stones

3.) Secondary hyperparathyroidism

- High PTH levels occur as a compensation for hypocalcemia; not a gland disease
- Commonly caused by vitamin D deficiency or chronic renal disease (since kidney cannot activate vitamin D)

4.) Rickets (children) and osteomalacia (adults)

- A disease of inadequate bone mineralization, usually caused by vitamin D deficiency
- Osteoblast produce adequate osteoid (matrix), but it fails to calcify, resulting in soft, weak bones

5.) Osteoporosis

- Most common bone disease in adults
- Not a problem with calcification but a problem in diminished organic bone matrix (less collagen); the osteoblastic activity is depressed relative to osteoclastic activity
- Common causes include inactivity, old age, and postmenopausal lack of estrogen (which normally stimulates OPG)

WEEK 11-15: INTEGRATIVE PHYSIOLOGY

SPORTS PHYSIOLOGY

MALE VS FEMALE ATHLETES

QUANTITATIVE DIFFERENCES

- Muscle strength and CO are 25% to 30% lower in female athletes compared to male athletes due to differences in body composition and muscle mass

STRENGTH PER AREA — $3\text{-}4/\text{cm}^2$

- Maximal contractile force of male & female is almost identical

HORMONAL EFFECTS

- Testosterone promotes a powerful anabolic effect for greater protein deposition and muscle mass (40% larger than female)
- Estrogen increase body fat deposition, detrimental in speed-based events but an advantage for insulation and buoyancy (e.g., channel swimming)

MUSCLE PERFORMANCE: STRENGTH, POWER & ENDURANCE

MUSCLE STRENGTH

- Determined by muscle size
- Holding strength (resisting a force that tries to stretch a contracted muscle) is about 40% greater than contractile strength (shortening the muscle)

MUSCLE POWER

- Measure of total work performed in a unit of time (e.g., kg-m/min)
- Body can produce extreme power for very short periods (8-10 seconds) but this drops to $\frac{1}{4}$ for long-term, endurance events

MUSCLE ENDURANCE

- Highly dependent on stored glycogen in the muscle before exercise
 - **High-carbohydrate diet:** 240 minutes of endurance
 - **Mixed diet:** 120 minutes of endurance
 - **High-fat diet:** 85 minutes of endurance

THE THREE ENERGY SYSTEMS

- The body has 3 primary metabolic systems to supply ATP (the energy currency) for muscle contraction

a.) Phosphagen system — anaerobic

- Uses stored ATP and phosphocreatine
- Provides maximal power for 8-10 seconds
- **Used for:** short events, all-out bursts like the 100-meter dash or weight lifting

b.) Glycogen-lactic acid system — anaerobic

- Breaks down muscle glycogen into lactic acid, creating ATP without oxygen
- Provides maximal power of 1.3 to 1.6 minutes
- **Used for:** intermediate events like the 400-meter dash or a 100-meter swim

c.) Aerobic system

- Oxidizes foodstuffs (fats, amino acids, glucose) in the mitochondria
- This is the slowest system at generating ATP but provides unlimited endurance (as long as nutrients last)
- **Used for:** long-duration endurance events, like a marathon

RECOVERY & NUTRIENT USE

a.) Oxygen debt

- The extra oxygen the body must consume after exercise to restore all systems
- Alactacid oxygen debt: restores the phosphagen system and stored oxygen in the lungs and blood
- Lactic acid oxygen debt: removes accumulated lactic acid (converts it back to glucose in the liver)

b.) Glycogen recovery

- Slow process; requires about 2 days on a high-carbohydrate diet for full recovery
- A high-fat/protein diet or fasting shows almost no glycogen recovery, even after 5 days

c.) Fuel source during exercise

- Carbohydrates are the nutrient of choice for intense activity
- In long endurance events (over 4-5 hours), glycogen stores are depleted, and the body shifts to using fat for 60-85% of its energy

ADAPTATIONS TO TRAINING

a.) Muscle hypertrophy

- Resistive training (contractions at $>50\%$ of max force) causes muscles to grow
- Increase in the diameter of muscle fibers

WEEK 11-15: INTEGRATIVE PHYSIOLOGY

b.) Fast-twitch vs. slow-twitch fibers

- Largely determined by genetics

- Fast-twitch: large, powerful fibers organized for anaerobic power (phosphagen and glycogen-lactic acid systems); used for **sprinting and jumping**
- Slow-twitch: smaller fibers with far more mitochondria and myoglobin; organized for aerobic endurance; used for **marathon running**

diminish performance; 5-10% can be serious

d.) Hyponatremia — low Na

- In endurance athletes, this is often caused by ingesting excessive amounts of hypotonic fluid (water), which dilutes the body's Na levels

e.) Acclimatization

- Over 1-2 weeks of heat exposure, the body (via aldosterone) adapts to conserve salt, and the sweat glands secrete sweat with a much lower salt concentration

CARDIOVASCULAR & RESPIRATORY SYSTEMS

1.) Respiration

- Not the most limiting factor in endurance performance
- Maximal breathing capacity is about **50% higher during maximal exercise**, providing a large safety margin
- During exercise, blood gases remain normal
- Respiration is stimulated by the signals from the brain and contracting muscles, not by a change in blood gas level

2.) Cardiovascular system

- Limiting factor for V O₂max (maximal oxygen utilization)
- Endurance training causes the heart to hypertrophy by up to 40%
- A trained athlete has a **high resting SV and low resting HR**
- During maximal exercise, trained heart has larger SV allows it to achieve a 40% greater maximal CO (important physiological benefit for an endurance athlete)

FITNESS AND LONGEVITY

- People who maintain fitness have mortality rates 3X lower than least fit (especially between ages 50-70)

BENEFITS

- Reduces CVD: lowers BP and improves cholesterol profiles
- Improves insulin sensitivity: reduces the risk of type 2 diabetes
- Provides bodily reserves: a higher “cardiac reserve” and “respiratory reserve” helps older, fit individuals survive illnesses like pneumonia
- Reduces cancer risk: lowers the risk for breast, prostate, and colon cancers

BODY HEAT & FLUIDS

a.) Heat production

- 75% to 80% of all energy used by the body is converted to heat
- Massive internal heat long during intense exercise

b.) Heatstroke

- If this heat cannot be eliminated, core body temperature rise to 106-108°F (41-42°C)
- Lethal condition that destroys cells, especially in the brain and requires immediate, rapid cooling

c.) Fluid loss

- Sweat loss as much as 3% of body weight can

REST-TO-EXERCISE TRANSITIONS

1.) Energy at rest

- At rest, the body is in homeostasis, and almost 100% of the ATP required is produced by aerobic metabolism

2.) ATP demand

- At the onset of exercise, the muscle's demand for ATP increases instantaneously

3.) O₂ uptake — V O₂

- O₂ consumption does not increase instantly
- It increases rapidly and reaches a steady state (where aerobic ATP production matches the demand) within 1-4 minutes

4.) Oxygen deficit

- This lag in O₂ uptake at the beginning of exercise is known as the O₂ deficit

5.) Anaerobic contribution

WEEK 11-15: INTEGRATIVE PHYSIOLOGY

- Because of this deficit, anaerobic energy sources must supply ATP at the start of exercise
- ATP-PC system is the first pathway activated, followed by glycolysis, and finally, aerobic metabolism

6.) Training and deficit

- Trained individuals have a better-developed aerobic capacity and reach a steady state faster, resulting in a small O₂ deficit

RECOVERY FROM EXERCISE

1.) EPOC — oxygen debt

- After exercise, metabolic rate remains elevated for several minutes (also called Excess Post-exercise O₂ Consumption)

“RAPID” COMPONENT OF EPOC

- Initial steep decline in V O₂ (first 2-3 minutes) is used to resynthesize stored ATP and phosphocreatine (PC) in the muscle and replenish O₂ stores in blood and tissues

“SLOW” COMPONENT OF EPOC

- The slower, more prolonged elevation in V O₂ is not primarily for lactate conversion
- Caused by:
 - Increased body temperature, which raises metabolic rate
 - Elevated levels of hormones (epinephrine and norepinephrine)
 - The energy needed for elevated HR and breathing
 - A portion of O₂ used for gluconeogenesis (lactate to glucose)

EPOC AND INTENSITY

- EPOC is greater following high-intensity exercise because of higher body temperature, greater PC depletion, and higher hormone levels

LACTATE METABOLISM

a.) Lactate removal

- Following exercise, lactate is not a waste product
 - ~70% is oxidized: it is converted to pyruvate and used as a fuel source by skeletal muscle and the heart
 - ~20% is converted to glucose: gluconeogenesis in the liver

- ~10% is converted to amino acids

b.) Active recovery

- Lactate removal is faster during light, continuous exercise (e.g., 35-50% V O₂ max) compared to a resting recovery, because the working muscles enhance its oxidation

c.) Lactate and soreness

- Lactate production does not cause delayed-onset muscle soreness (DOMS)
- DOMS is likely caused by microscopic injury to muscle fibers

METABOLIC RESPONSES TO EXERCISE DURATION AND INTENSITY

1.) Short-term, intense — 2-20 seconds

- ATP production is dominated by the ATP-PC system

2.) Intense — >20 seconds

- Anaerobic glycolysis

3.) Events > 45 seconds

- Combination of all 3 systems (ATP-PC, glycolysis, aerobic)
- An event lasting 2-3 minutes is roughly a 50/50 split between anaerobic and aerobic production

4.) Prolonged — >10 minutes

- Aerobic metabolism

5.) Upward VO₂ drift

- During prolonged exercise, a steady state may not be maintained
- Oxygen uptake can “drift” upward if exercise is performed in a hot/humid environment or at a high intensity (>75% VO₂ max)

INCREMENTAL EXERCISE CONCEPTS

1.) VO₂ max — maximal O₂ uptake

- The “*physiological ceiling*” for the body’s ability to transport and utilize oxygen during exercise
- Considered the most valid measurement of cardiovascular fitness
 - **Reduced rate of lactate removal from the blood**

WEEK 11-15: INTEGRATIVE PHYSIOLOGY

2.) Lactate threshold — LT

- Point during an incremental exercise test where blood lactate levels begin to rise exponentially
- Occurs at a lower % of VO₂ max in untrained individuals (50-60%) compared to trained individuals (65-80%)
- Causes:
 - Low muscle oxygen (hypoxia)
 - Accelerated glycolysis (e.g., from epinephrine) produces NADH faster than the mitochondria can shuttle it, forcing pyruvate to accept the H and become lactate
 - Recruitment of fast-twitch muscle fibers: contain an enzyme (LDH isozyme) that favors the formation of lactate
 - A reduced rate of lactate removal from the blood

FUEL UTILIZATION DURING EXERCISE

1.) Respiratory exchange ratio — R

- Ratio of CO₂ produced to O₂ consumed (VCO₂/VO₂)
- Used to estimate fuel utilization only during steady-state exercise
 - An R of 0.70 indicates 100% fat metabolism
 - An R of 1.00 indicates 100% carbohydrate metabolism
 - An R of 0.85 indicates a 50/50

2.) Fuel shift — duration

- During prolonged, moderate-intensity exercise, there is a gradual shift from carbohydrate to fat metabolism
- This is due to rising epinephrine levels over time stimulate lipolysis (the breakdown of triglycerides), increasing the availability of free fatty acids (FFAs) as fuel

3.) “Fats burn in the flame of carbohydrates”

- Depletion of carbohydrate stores (glycogen) can cause muscular fatigue
- When glycogen is depleted, the rate of glycolysis slows, reducing the amount of pyruvate; slows down the Krebs cycle, as its intermediates are reduced
- Because fat is metabolized during Krebs cycle, a lack of carbohydrates (and the resulting slowdown of the Krebs cycle) reduces the rate of fat metabolism

FUEL SOURCES

CARBOHYDRATE SOURCES

a.) Muscle glycogen

- Primary carbohydrate source during high-intensity exercise and the first hour of prolonged exercise

b.) Blood glucose — from liver glycogen

- Primary source during low-intensity exercise; increases as muscle glycogen is depleted during prolonged exercise

FAT SOURCES

a.) Muscle triglycerides

- Used more as exercise intensity increases, contributing about equally with plasma FFAs at 65-85% VO₂ max

b.) Plasma free fatty acids (FFAs)

- Primary source of fat during low-intensity, prolonged exercise

LACTATE SOURCES

a.) Cori cycle

- Lactate produced in the muscle is transported to the liver, where it is converted back to glucose (gluconeogenesis) and returned to the blood

b.) Lactate shuttle

- Lactate is not a waste product
- It can be produced in one tissue (e.g., fast-twitch fibers) and transported to another (e.g., slow-twitch fibers or the heart) to be used directly as fuel after being converted back to pyruvate

THE “CROSSOVER” CONCEPT — INTENSITY

- “**Crossover point**”: intensity at which the body shifts from primarily using fat to primarily using carbohydrate
- This shift is caused by the recruitment of fast-twitch fibers (which prefers carbs) and rising epinephrine levels (which stimulates glycogen breakdown)

a.) Low-intensity exercise (e.g., <30% VO₂ max):

fat is the primary fuel source

b.) High-intensity exercise (e.g., >70% VO₂ max):

carbohydrate primary fuel source

EXERCISE AND MUSCLE FATIGUE

- Fatigue is a reduction in muscle power output, resulting from decreased force generation or shortening velocity

WEEK 11-15: INTEGRATIVE PHYSIOLOGY

a.) High-intensity exercise — 1 to 10 mins

- Caused by:

- H ions
- Inorganic phosphate (Pi)
- Free radicals

- These interfere with Ca sensitivity (e.g., H ions compete with Ca for binding sites on troponin) and inhibit cross-bridge function

b.) Prolonged, moderate-intensity exercise — > 1 hour

- Caused by depletion of muscle glycogen and accumulation of free radicals

- Role: ideal for explosive, short-term power events (e.g., sprinting, weightlifting)

3.) Type IIa — intermediate

- A mix of type I and type IIx properties; fast-twitch but also have a moderate aerobic capacity

- Performance: power athletes (sprinters) have a high percentage of fast fibers, while endurance athletes (marathoners) have a high percentage of slow-fibers; **average person has roughly a 50/50 mix**

PRIMARY GOAL AND ADJUSTMENTS OF CARDIORESPIRATORY

CORE PURPOSE

- To deliver adequate O₂ and nutrients, remove wastes, and help regulate body temperature

CHALLENGES OF EXERCISE

- During heavy exercise, the muscle's demand for O₂ can increase 15-25X compared to rest

TWO MAJOR ADJUSTMENTS

- a.) Increase CO:** heart pumps more blood per minute
b.) Redistribution of blood flow: blood is shunted away from inactive organs and toward the active skeletal muscles

CRAMPS EXERCISE-ASSOCIATED MUSCLE CRAMPS

a.) Involuntary contractions

- Cramps are spasmodic, involuntary muscle contractions

b.) CNS theory — most likely cause

- Cramps are not caused by dehydration or electrolyte imbalance, but by **abnormal activity in the CNS**
- Muscle spindles (detects stretch) become hyperexcitable
- Golgi tendon organs (GTOs), which detect tension, becomes inhibited
- Result: net **increase in motor neuron firing from the SC**, leading to cramps
- Intervention: **passive stretching to activate GTO**, to send a strong inhibitory signal to the motor neuron

MUSCLE FIBER TYPES

1.) Type I — slow-twitch

- Metabolism: high mitochondria, myoglobin, capillary density; highly aerobic and fatigue-resistant
- Contractile: low myosin ATPase activity (slow contraction speed), highly efficient, lower force production
- Role: ideal for endurance events (e.g., marathon running)

2.) Type IIx — fast-twitch

- Metabolism: low mitochondria, rich in glycolytic enzymes; primarily anaerobic and fatigues quickly
- Contractile: highest myosin ATPase activity (fastest contraction speed), least efficient, highest power and force

EXERCISE

EXERCISE CARDIOPROTECTION

- Regular endurance exercise is cardioprotective
- Causes biochemical changes in the heart that can reduce the amount of cardiac muscle damage by ~60% if a heart attack occurs

EFFECT OF EXERCISE

- As HR increases, the time spent in both systole and diastole decreases, but the **greatest time reduction occurs during diastole**
- This means there is less time for ventricles to fill with blood

REGULATION OF CO — HR & SV

HR REGULATION

- a.) Parasympathetic NS:** acts as the “brake” via the

WEEK 11-15: INTEGRATIVE PHYSIOLOGY

vagus nerve; at rest, this system dominates, keeping HR low

b.) Sympathetic NS: Acts as the “accelerator.” increasing HR and the force of contraction

c.) During exercise: The initial rise in HR (~100 bpm) is due to parasympathetic withdrawal (taking the brake off); further increases are caused by increased sympathetic stimulation (hitting the gas)

CIRCULATORY RESPONSES TO EXERCISE

1.) O₂ delivery

- VO₂ increases during exercise due to increase in both CO (Q) and the a-VO₂ difference (muscles get better at extracting O₂)

2.) SV responses

- In untrained individuals, SV plateaus at 40-60% VO₂ max; from that point on, all increases in CO from increasing HR

- In highly trained endurance athletes, SV does not plateau and continues to increase all the way to VO₂ max

3.) Blood flow redistribution

- At rest, muscles receive ~15-20% of CO

- ***During maximal exercise, muscles receive 80-85% of CO***

4.) Autoregulation (local control): redistribution is controlled locally

- Active muscles produce metabolic byproducts (like NO, adenosine, K); ***these are powerful vasodilators***

- This local vasodilators overrides the body's global sympathetic signal to vasoconstrict (sympatholysis), shunting blood precisely where it's needed most

5.) Arm vs. leg exercise

- At the same O₂ uptake, arm exercise causes a higher HR and BP than leg exercise

- This is due to arm exercise involving a smaller muscle mass leading to more vasoconstriction in the large, inactive leg muscles, which raises total vascular resistance

6.) Cardiovascular drift — prolonged exercise

- Prolonged, steady-state exercise (especially in the heat, CO is maintained)

- However, due to sweating, plasma volume increases, which reduces VR and thus SV gradually decreases

- To compensate and maintain CO, the HR gradually increases (cardiovascular drift)

PRINCIPLES OF TRAINING

1.) Overload

- To achieve adaptation, an organ system or tissue must be exercised at a level beyond which it is accustomed

- Typically manipulated through intensity, duration, and frequency

2.) Reversibility

- Fitness gains are quickly lost when training is stopped and the overload is removed

3.) Specificity

- Training adaptations are highly specific to the muscles involved, the fiber types recruited, the primary energy system used, the velocity of contraction, and the type of contraction (concentric or eccentric)

- **Endurance training** primarily increases mitochondria and capillaries
- **Resistance training** primarily increases contractile proteins

ENDURANCE TRAINING AND VO₂ MAX

1.) VO₂ max

- Maximal capacity of the body to transport and utilize O₂ during dynamic exercise

- Training response:

- Endurance training (e.g., 20+ min, 3+ times/week, >50% VO₂ max) typically increases VO₂ max by 15-20% over 2-3 months

- Genetic influence:

- Play a major role
- Heritability **determines about 50% of initial VO₂ max**
- The magnitude of improvement from training is also highly genetic (about 47% heritable), leading to “high responders” (who can see up to 50% improvement) and “low responders” (who may only see 2-3% improvement)

CARDIOVASCULAR ADAPTATIONS TO ENDURANCE TRAINING

- Increase in VO₂ is explained by the Fick equation:

- $VO_2 \text{ max} = \text{max CO} \times \text{Max a} - VO_2 \text{ difference}$

SHORT-TERM TRAINING — 1 TO 4 MONTHS

- The VO₂ max increase is due entirely to an increase in maximal CO

WEEK 11-15: INTEGRATIVE PHYSIOLOGY

LONG-TERM TRAINING — MONTHS TO YEARS

- The VO₂ max increase due to increase in both maximal CO and maximal a - VO₂ difference

MAXIMAL HR

- Does not increase with training and may even decrease slightly
- Therefore, all increases in CO are due to SV

INCREASED MAXIMAL SV

- Primary adaptation in CO and is caused by:

a.) Increased preload — EDV

- The ventricle fills with more blood before contracting due to:

- Rapid increase in plasma volume
- Increased ventricular filling (from slower resting HR, or bradycardia)
- Increased ventricular volume

b.) Increased contractility

- The heart contracts more forcefully, likely due to improved “twist mechanics” and sympathetic NS adaptations

c.) Decreased afterload

- Trained muscles have less vascular resistance, making it easier for the heart to pump blood out

INCREASED MAXIMAL VO₂ DIFFERENCE

- This adaptation means the muscles become better at extracting O₂ from the blood due to:

- **Increased capillary density (primary):** more capillaries in the muscle mean blood flows slower, allowing more time for O₂ to diffuse from the blood into the muscle fiber
- **Increased mitochondrial volume (secondary):** the muscle’s capacity to use oxygen increases, but the body’s ability to deliver oxygen is usually the limiting factor for VO₂ max

SKELETAL MUSCLE ADAPTATIONS TO ENDURANCE TRAINING

- Endurance training improves submaximal performance by making the muscle more efficient and better at maintaining homeostasis

FAST-TO-SLOW SHIFT

- Training causes a shift from fast-twitch fibers (type IIx) toward more efficient, fatigue-resistant fibers (type IIa and type I)
- This allows for **more work to be done with less ATP used**

INCREASED MITOCHONDRIAL CONTENT

- Training (both continuous and HIIT) can **increase mitochondrial volume by 50-100% within 6 weeks**
- Training also increases mitochondrial turnover (mitophagy), replacing old, damaged mitochondria with new, healthy ones

METABOLIC CONSEQUENCE OF MORE MITOCHONDRIA

- For any given submaximal workload, the ATP demand is the same, but this work is now shared by more mitochondria
- This means a lower [ADP] concentration is needed in the cytosol to stimulate ATP production
- This lower [ADP] results in a small O₂ deficit (reach steady-state faster), less phosphocreatine (PC) depletion, and less stimulation of glycolysis

“GLYCOGEN SPARING” — INCREASED FAT METABOLISM

- Trained muscles decrease their use of glucose and increase their use of fat as fuel

- Mechanism:

- **More fat delivery:** increased in capillary density
- **More fat transport:** more fatty acid transporters (FAT) move fat from the blood into the muscle and from the cytoplasm into the mitochondria
- **More fat oxidation:** more mitochondria mean more beta-oxidation enzymes to break down fat into acetyl-CoA
- **Feedback:** the resulting high levels of citrate (from the Krebs cycle) inhibit PFK, the rate-limiting enzyme for glycolysis; this inhibition spares carbohydrates

IMPROVED ACID-BASE BALANCE

- Trained muscles produce less lactate and H⁺ at any given submaximal workload

WEEK 11-15: INTEGRATIVE PHYSIOLOGY

- Mechanism: less pyruvate is formed (due to glycogen sparing), and more "NADH shuttles" transport NADH into mitochondria, preventing it from reacting with pyruvate to form lactate

IMPROVED ANTIOXIDANT CAPACITY

- Training increases the muscle's endogenous (self-made) antioxidant enzymes, which protect against free radical damage and fatigue

DETRAINING — ENDURANCE

- Rapid decline: VO₂ max drops quickly
- Initial drop (first 12 days): caused by a decrease in maximal SV, which is due to a rapid loss of plasma volume
- Later drop (>21 days): caused by a decrease in the a - VO₂ difference, which is due to the loss of mitochondria
- Mitochondrial loss: lost very fast; about 50% of mitochondrial gain from training is lost after just 1 week of detraining

ANAEROBIC TRAINING

- Definition: short (10-30 seconds), all-out, high-intensity efforts
- Performance: increases peak anaerobic power by 3-25%
- Muscle adaptations:
 - Improves muscle buffering capacity
 - Causes hypertrophy of fast (type II) fibers
 - Increases enzymes in the ATP-PC and glycolic systems
 - High-intensity interval training (HIIT > 30s) can also promote mitochondrial progress

PHYSIOLOGICAL EFFECTS OF RESISTANCE TRAINING

a.) Muscular strength

- The maximal force a muscle can generate, often measured as a one-repetition maximum (1-RM)

b.) Muscular endurance

- Ability to make repeated contractions against a submaximal load

c.) Training specificity

- Training with heavy weights (6-10 reps) primarily increases strength, while training with lighter weights (35-40 reps) primarily increases muscular endurance

NEURAL ADAPTATIONS VS. MUSCLE GROWTH

- The increase in strength from resistance training is a combination of NS adaptations and physical changes in the muscle

a.) Short-term gains — first 2-8 weeks

- The rapid increase in strength at the beginning of training is primarily due to neural adaptations
- Strength increases significantly before any major muscle hypertrophy occurs

b.) Long-term gains

- Further increases in strength are dominated by muscle hypertrophy and increases in fiber specific force production

NEURAL ADAPTATIONS TO RESISTANCE TRAINING

INCREASED NEURAL DRIVE

- Primary neural adaptation which is the magnitude of the electrical signal sent from the CNS to muscle

- **Achieved by:**

- Reduced neural inhibition
 - Resistance training desensitizes GTO
 - GTO normally acts as a "safety brakes" to prevent excessive force
 - By reducing inhibiting signal, the CNS can activate motor units more effectively
- **Increased motor unit synchronization**
 - Resistance training causes more motor units to fire at the same time, leading to a more forceful contraction
- **Cross-education**
 - Training one limb results in strength gains in the untrained contralateral limb
 - This phenomenon is clear proof of the NS's role in strength

MUSCLE ADAPTATIONS TO RESISTANCE TRAINING

1.) Hypertrophy

- Primary way muscle mass increases with long-term training

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- Results from an increase in myofibrillar proteins (actin and myosin), which adds more sarcomeres in parallel, making the fiber thicker and stronger
- Type II (fast-twitch) fibers show a greater degree of hypertrophy than type I fibers

2.) Hyperplasia

- Observed in animals, likely through fiber splitting
- In humans, it remains highly controversial; not considered as a major contributor to muscle growth

3.) Fiber specific tension

- Resistance training can increase the specific force production (force per cross-sectional area) of Type I fibers only
- Thought to be from an increase Ca sensitivity

4.) Fiber type shift

- Resistance training causes a fast-to-slow shift, specifically a ***conversion of type Ix fibers to type Ila fibers***
- It does not increase the % of type I fibers

5.) Other adaptations

- Antioxidant enzymes: resistance training improves the muscle's antioxidant capacity (e.g., ~100% increase in 12 weeks)
- Connective tissue: training increases the strength and size of tendons and ligaments in parallel with muscle strength gains
- Bone mineral content: training increases bone density, making bone stronger

HOW MUSCLE GROWTH OCCURS

1.) Protein synthesis

- Muscle growth occurs when the rate of muscle protein synthesis is greater than the rate of protein breakdown
- ***A single bout of resistance increases protein synthesis by 50-150% within 1-4 hours post-exercise***
- This response lasts longer in untrained individuals, which is why they initially see faster gains

2.) Roles of hormones

- Resistance training-induced hypertrophy does not require an increase in circulating hormones
- Testosterone and IGF-1 can support growth

- ***The primary signal is the mechanical stretch within the muscle***

3.) Genetic influence

- Genetic play a huge role; can be “high-responders,” “moderate responders,” or “nonresponders” to training, largely due to genetic differences in their ability to activate protein synthesis and satellite cells

DETRAINING AND INACTIVITY

- Leads to a loss of strength and muscle atrophy
- Loss of strength is slower than the loss of endurance

a.) Strength of loss is primarily neural at first

- After 20 weeks of detraining, strength loss (~31%) is far greater than the loss of muscle fiber size (~2-14%), suggesting the initial loss is due to a decline in neural drive

b.) Muscle memory

- Upon retraining, strength and muscle size are regained much faster than they were gained initially
- A theory suggests this because the myonuclei gained during the initial training persist during detraining, providing a permanent “blueprint” for rapid regrowth

c.) Inactivity — atrophy

- Prolonged inactivity (like bed rest or casting) causes rapid muscle atrophy (e.g., 7-10% loss in 7 days)
- Mechanism: atrophy is caused by both decrease in protein synthesis and an increase in protein breakdown
- Key trigger: trigger for disuse atrophy increase free radical production (oxidative stress) in the inactive muscle, which inhibits mTOR (stopping synthesis) and activates proteases (starting breakdown)

CONCURRENT STRENGTH AND ENDURANCE TRAINING

THE INTERFERENCE EFFECT

- Performing endurance and strength training on the same day can impair strength gains compared to doing strength training alone
- When it occurs: most likely when the endurance training is high in volume (>30 min/session) and

WEEK 11-15: INTEGRATIVE PHYSIOLOGY

frequency (>2 days/week)

MOLECULAR MECHANISM — THE CONFLICT

- **Resistance training activates mTOR** -> promotes protein synthesis and muscle growth
- **Endurance training activates AMPK** -> promotes mitochondrial biogenesis
- The problem: **AMPK activation inhibits MTOR**, which means the endurance signal can biochemically block the muscle-building signal from the resistance session